

William R. Arida

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EDUCATION

The Ohio State University

B.S. Mechanical Engineering, Robotics Minor

May 2026

QUALIFICATIONS

Coursework

- Machining, System Integration, Machine Elements, Measurements, Manufacturing, Kinematics, Robotic Mechanisms, Sustainable Energy, Machine Learning, Real Time Robotics, Multidisciplinary Capstone

Programming & Software

- SolidWorks, MATLAB, LabVIEW, Arduino, Machining (lathe, milling, boring), Python, C++

Experience With

- Mechanical prototyping, component selection, assembly, electromechanical integration, documentation

EXPERIENCE

CRP Industries

May 2025 - August 2025

Product Development Assistant Internship

Cranbury, NJ

- Supported head of product development in identifying new product line opportunities in automotive aftermarket
- Researched part failure patterns to recommend bundled offerings commonly replaced together during auto repairs
- Forecasted opportunities for electrical components entering the aftermarket based on their lifecycle
- Tracked supplier outreach and competition research using Excel

Valcor Engineering Corp.

May 2024 - August 2024

Production Testing Support Engineer Internship

Springfield, NJ

- Organized and rewrote testing documentation for Aerojet fill/drain valve
- Constructed and implemented high pressure solenoid actuation testing on Aerojet valve
- Assembled valves from multiple client programs across Valcor's product line
- Assisted technicians in solenoid actuation testing and Blue Origin quick disconnect valve test setup

RELEVANT PROJECTS

CBI Rat Testing Chamber | Multidisciplinary Capstone

August 2025 - May 2026

- Translated neuroscience laboratory needs into mechanical and electromechanical design requirements
- Designed the internal-door mechanism in SolidWorks and selected materials, motors, sensors, and components
- Developed door controls and helped fabricate, assemble, integrate, and test the completed chamber

Robotic Gripper Arm

January 2025 - May 2025

- Designed and built a 4-DOF robotic arm for object manipulation and path-following
- Modeled the mechanism in SolidWorks and fabricated custom components using 3D printing
- Integrated servo and stepper motors with Arduino using forward and inverse kinematics

Air Motor Piston Fabrication Project

March 2024 - May 2024

- Machined piston components using lathe, drilling, and milling techniques based on given designs
- Applied precision measuring and machining skills, working within project tolerances

Automated House Plant Irrigation System

January 2024 - March 2024

- Designed and built automated Arduino-based watering system with real-time moisture sensing
- Planned circuit layout, programmed control, fabricated housing, and documented testing

LEADERSHIP AND INVOLVEMENT

Lunabotics Club

Autumn 2024 - May 2025

- Mechanical team member, collaborated in claw and arm design for University Rover Challenge 2025

Sigma Epsilon Phi (Greek cultural heritage club)

Autumn 2020 - Present

- Participated in cultural and volunteer activities such as food drives and community fundraisers

OCF Club (Greek Orthodox Fellowship)

Autumn 2024 - Present

- Engaged in faith based discussions and organized community service projects

Eagle Scout Award and Project:

July 2020

- Led church landscaping and infrastructure project, revitalizing courtyards, planting, mulching, installing benches